

MTH212M ASSIGNMENT 1

Instructions:

- (i) Each question is of 10 marks.
- (ii) Only one question will be evaluated. The question number (to be evaluated) will be declared after the deadline for submission.
- (iii) Deadline for submission: **10:00 am IST, Friday January 30, 2026.**
- (iv) **Assignment submission will be through the HelloIITK portal.** Email submissions and any late submission through the HelloIITK portal will NOT be graded.
- (v) Please **upload one single PDF file** containing all your answers.
- (vi) Provide all details/justifications.

Question 1. Consider a Markov chain $\{X_n\}_{n=0}^{\infty}$ with state space $S = \mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ and the transition probabilities are given by

$$\mathbb{P}(X_{n+1} = j | X_n = i) = \begin{cases} \frac{1}{2}, & \text{if } j = i, \\ \frac{1}{4}, & \text{if } j = i + 1, \\ \frac{1}{4}, & \text{if } j = i - 1, \\ 0, & \text{otherwise,} \end{cases}$$

for $i \in \mathbb{Z}$.

- (a) (3 marks) Find the communicating classes.
- (b) (2 marks) Find the periods of all the states.
- (c) (5 marks) Check the recurrence/transience behaviour of all the states.

Question 2. Fix $p \in (0, 1)$. Consider a Markov chain $\{X_n\}_{n=0}^{\infty}$ with state space $S = \{0, 1, 2, \dots\}$ and the transition probability matrix P as follows:

$$\begin{pmatrix} p & 1-p & 0 & 0 & 0 & 0 & \dots \\ p & 0 & 1-p & 0 & 0 & 0 & \dots \\ p & 0 & 0 & 1-p & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- (a) (3 marks) Find the communicating classes.
- (b) (2 marks) Find the periods of all the states.
- (c) (4 marks) Check the recurrence/transience behaviour of all the states.
- (d) (1 marks) Compute $\lim_{n \rightarrow \infty} p_{00}^n$.